Work along a displacement

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate work and identify when a force does no work.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 13; 3 course days later.*
Shift 15 m. Assumes a constant added transverse velocity and a straight-line comparison.

02. Explain a worked example (worked)
Parallel arrows; rectangle 5 N high and 3 m wide, area 15 J.

03. Look and explain (model)
Both right; positive work.

04. Build the solution (practice)
40 J; 20 J.

05. Use the idea (reason)
No displacement so zero mechanical work on bag. Muscles can still use internal chemical energy.

06. Choose a method (mixed) *Review: Unit 3, Day 4; 12 course days later.*
The new force magnitude is 1/4 of the original because the same momentum change occurs over 4 times the time.

07. Exit check (exit)
15 J.

08. STRETCH: Apply it another way (stretch)
Examples 4 N over 5 m and 10 N over 2 m; force parallel to displacement.

Work along a displacement

antocciphysics | TEACHER KEY | HONORS

Learning target: Calculate work and identify when a force does no work.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 13; 3 course days later.*
Shift multiplier 5/3≈1.667 because shift=(J/M) t. Assumes constant added transverse velocity.

02. Explain a worked example (worked)
Parallel arrows; rectangle 5 N high and 3 m wide, area 15 J.

03. Represent and interpret (model)
Both right; positive work.

04. Solve and justify (practice)
+90 J,-30 J, net +60 J.

05. Analyze the reasoning (reason)
No displacement so zero mechanical work on bag. Muscles can still use internal chemical energy.

06. Choose a method (mixed) *Review: Unit 3, Day 4; 12 course days later.*
J=-16 N·s; first F≈-10.67 N; longer-time F≈-5.333 N. Both oppose the initial velocity.

07. Exit check (exit)
24 N; 0 J.

08. STRETCH: Apply it another way (stretch)
Trapezoid area (2+8)/2×4=20 J; using final force gives 32 J and overestimates.